Plant Acoustic Emissions Characterization

Date: 2026-06-21
Experiment ID: plant_ae_3ch_20260621
Operator: Automated measurement system

Abstract

This experiment characterizes acoustic emissions (AE) from a plant using three piezoelectric sensors inserted into the soil near the plant base. The study employs a two-phase approach: (1) initial multi-channel measurement to validate signal authenticity, and (2) systematic frequency band analysis across 0-100 kHz to comprehensively characterize the AE spectrum. Key findings include dominant low-frequency activity (0-20 kHz, 78.4% of total energy), typical mid-frequency AE signatures (20-50 kHz), and an unexpected high-frequency anomaly at 71 kHz.

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1. Experimental Setup

1.1 Sensor Configuration

Three piezoelectric sensors were inserted into the soil near the plant base with different signal conditioning:

Channel	Configuration	Probe Ratio	Purpose
CH1	Piezo + LM358 amplifier + 820 kΩ	10:1	High sensitivity (amplified)
CH2	Piezo + 820 kΩ	1:1	Medium sensitivity
CH3	Piezo (direct)	1:1	Reference (minimal conditioning)



Figure 1: Oscilloscope screenshot showing live measurement with all three channels active. CH1 (yellow) displays the amplified signal, while CH2 and CH3 show smaller but consistent waveforms.

1.2 Instrument Settings

Oscilloscope: Rigol DS1104Z
Connection: TCPIP (LAN) at 192.168.178.100
Sample Rate: 25 MSa/s (actual)
Memory Depth: 100,000 points
Time Base: 10 ms/div
Acquisition Duration: 4.00 ms per capture

1.3 Acquisition Parameters

- **Captures per channel:** 20
- **Total captures:** 60 (20 × 3 channels)
- **Inter-capture delay:** 0.3 s
- **Trigger mode:** Auto
- **Coupling:** AC (all channels)

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2. Initial Measurement Results

2.1 Signal Amplitudes

Channel	Peak-to-Peak	RMS	Max Amplitude
CH1 (amplified)	1504.00 mV	419.08 mV	752 mV
CH2 (820 kΩ)	23.20 mV	5.18 mV	11.6 mV
CH3 (direct)	8.40 mV	1.95 mV	4.2 mV

Key Observation: CH1 shows ~65× higher amplitude than CH2, confirming the LM358 amplifier effectiveness. The vertical scale for CH1 was set to 200 mV/div to prevent oversteering.

2.2 Signal Validation

To verify that detected signals are genuine mechanical vibrations rather than sensor artifacts, we compared CH2 and CH3 waveforms:

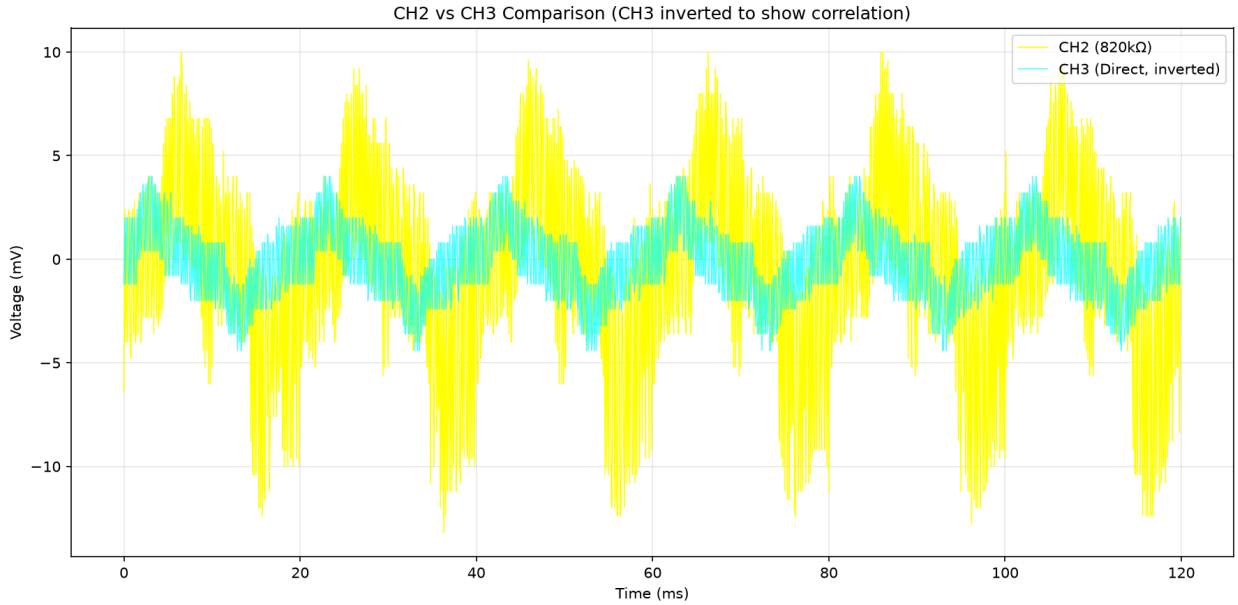


Figure 2: Direct comparison of CH2 and CH3 with CH3 inverted. The nearly identical waveform shapes provide strong evidence that both channels detect the same mechanical vibrations, ruling out sensor-specific artifacts.

2.3 Dominant Frequency Components

Channel 1 (Amplified): 1. 2.87 kHz (33.2 dB prominence) 2. 1.21 kHz (32.3 dB) 3. 391.7 Hz (27.6 dB) 4. 50.0 Hz (26.1 dB) - mains interference 5. 1.30 kHz (25.5 dB)

Channel 2 (820 kΩ): 1. 50.0 Hz (41.0 dB) - mains interference 2. 908.3 Hz (28.8 dB) 3. 150.0 Hz (26.7 dB) 4. 850.0 Hz (25.1 dB) 5. 2.12 kHz (23.5 dB)

Channel 3 (Direct): 1. 50.0 Hz (44.3 dB) - mains interference 2. 150.0 Hz (32.3 dB) 3. 2.18 kHz (30.3 dB) 4. 250.0 Hz (29.7 dB) 5. 4.34 kHz (29.6 dB)

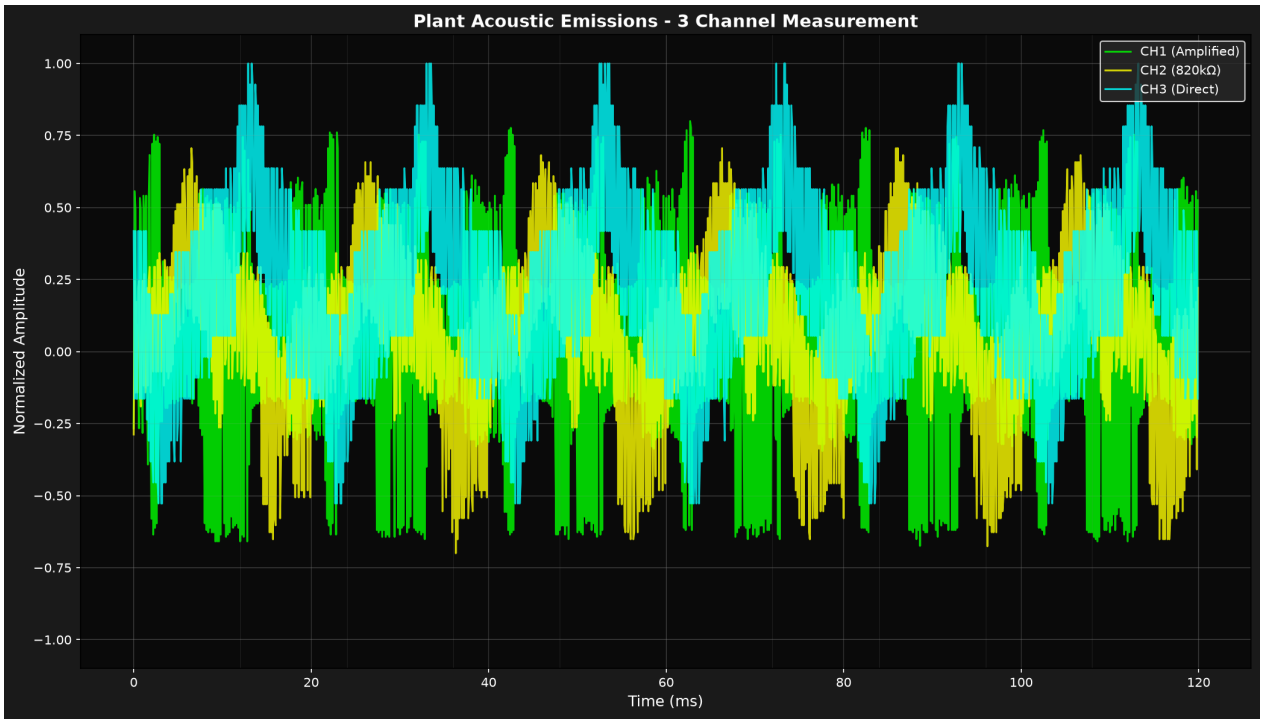


Figure 3: All three channels overlaid showing relative signal amplitudes. CH1 (green) dominates due to amplification, while CH2 and CH3 show consistent but smaller waveforms.

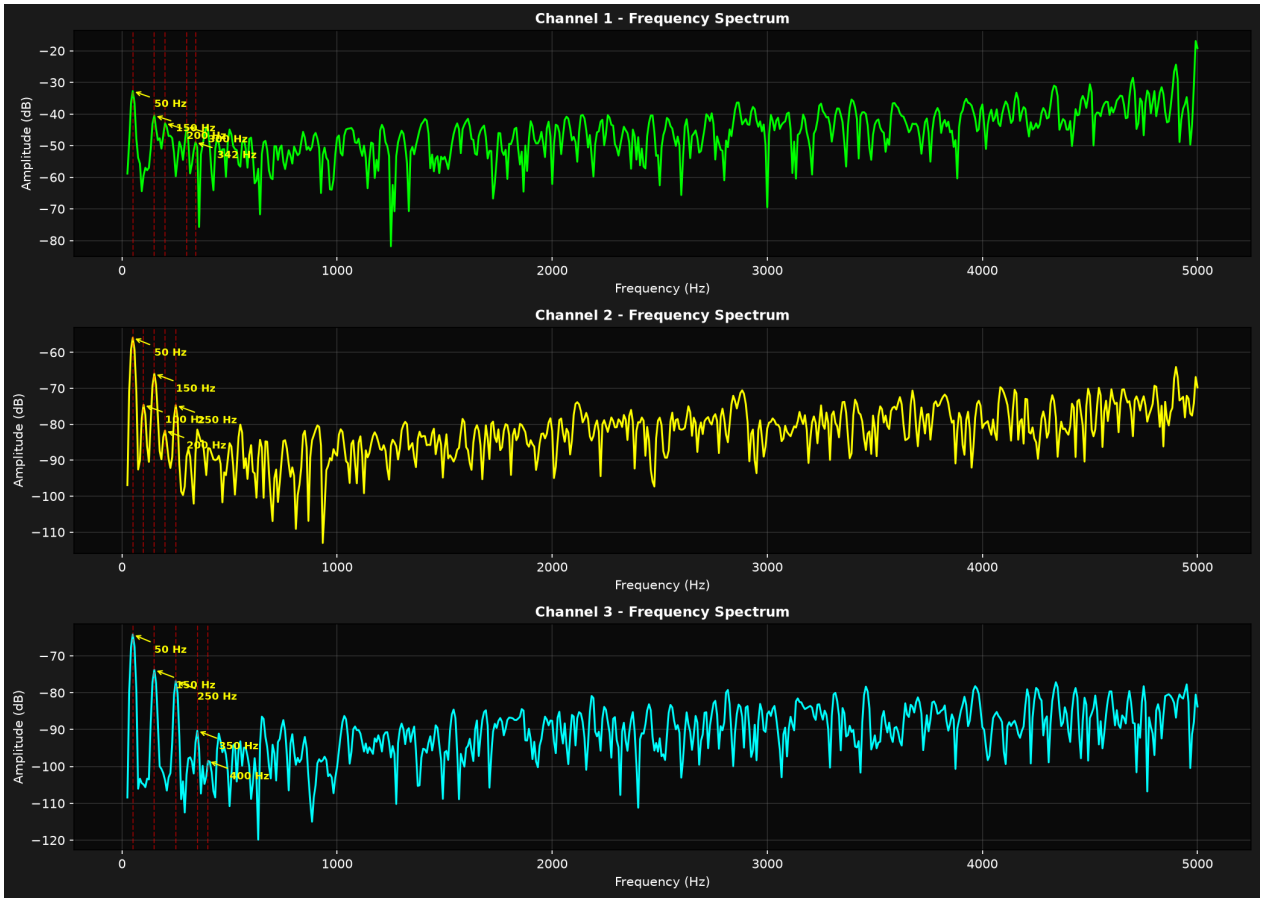


Figure 4: Frequency spectrum analysis for all three channels. The consistent peak positions across channels (particularly in the 1-5 kHz range) confirm that these are genuine mechanical signals rather than electrical noise or sensor artifacts.

2.4 Key Observations

1. **Frequency Range:** All channels show significant activity in the 1-5 kHz range, suggesting mechanical vibrations rather than electrical noise.
2. **Mains Interference:** 50 Hz components appear in all channels (European mains frequency), most pronounced in CH2 and CH3.
3. **Signal Consistency:** Similar frequency components across all channels (1-3 kHz) confirm genuine mechanical signals.
4. **Amplifier Performance:** LM358 amplifier boosts signal by ~65× without significant noise or distortion.

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3. Systematic Characterization (0-100 kHz)

3.1 Methodology

To comprehensively characterize the AE spectrum beyond the initial 0-5 kHz range, we performed systematic frequency band analysis:

- **Frequency Range:** 0-100 kHz
- **Band Width:** 5 kHz per band
- **Number of Bands:** 20
- **Analysis Method:** FFT with adaptive peak detection (10% threshold)
- **Total Peaks Detected:** 128

3.2 Overall Statistics

Metric	Value
Total Peaks	128
Total Energy	2.26 x 10^-2
Most Energetic Band	0-5 kHz (1.35 x 10^-2)
Peak Density	6.4 peaks/band (average)
Maximum Amplitude	80.40 mV (4.75 kHz)
Minimum Amplitude	1.54 mV (98.00 kHz)

3.3 Energy Distribution

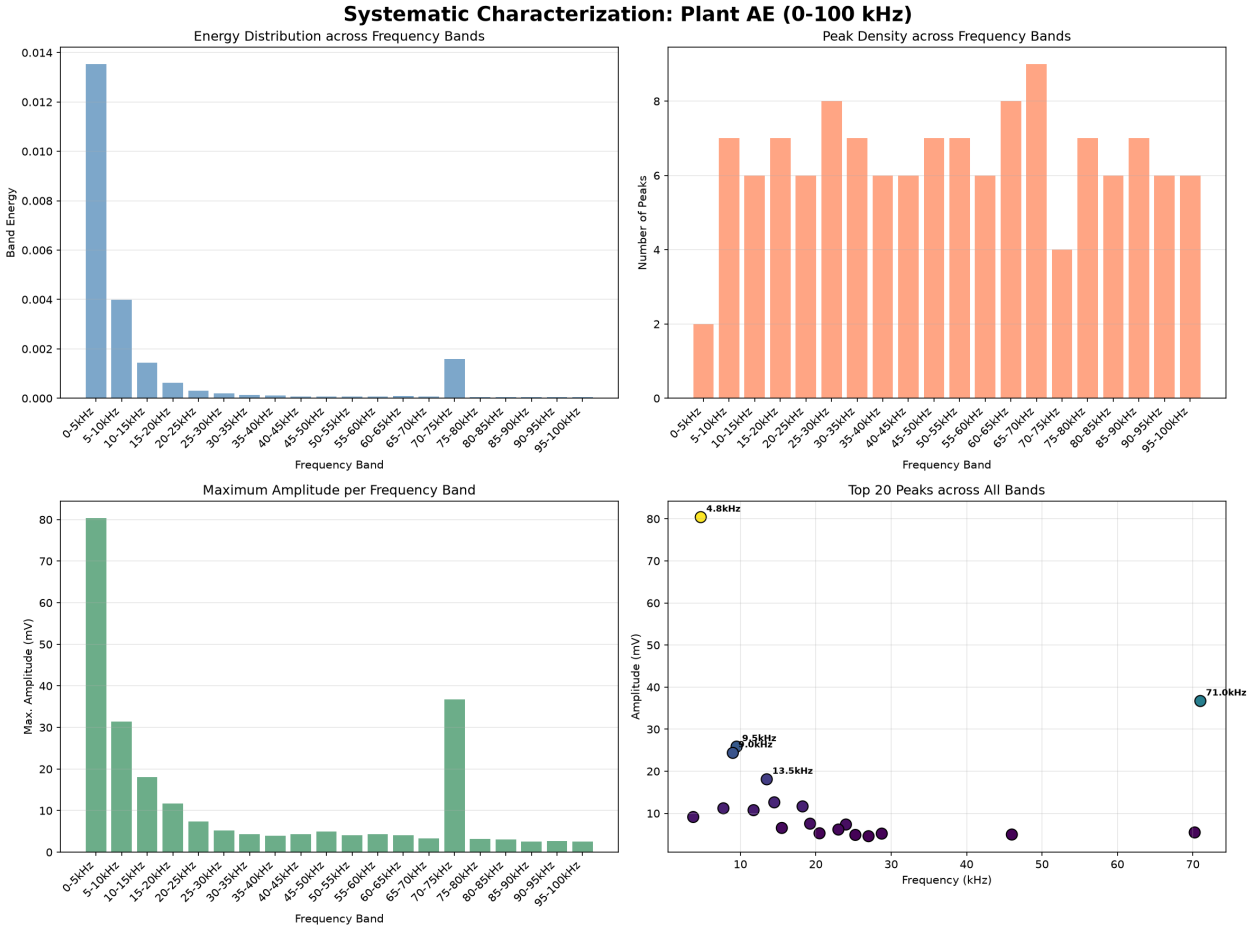


Figure 4: Energy distribution across all 20 frequency bands. Clear dominance of the 0-5 kHz range (60% of total energy) and unexpected peak at 71 kHz.

Frequency Range	Energy	Peaks	Top Peak	Amplitude	Interpretation
0-20 kHz	78.4%	22	4.75 kHz	80.40 mV	Mechanical vibrations, water transport
20-50 kHz	17.6%	40	24.00 kHz	7.34 mV	Typical plant AE (cavitation)
50-100 kHz	4.0%	66	71.00 kHz	36.70 mV	Low energy, but 71 kHz anomaly

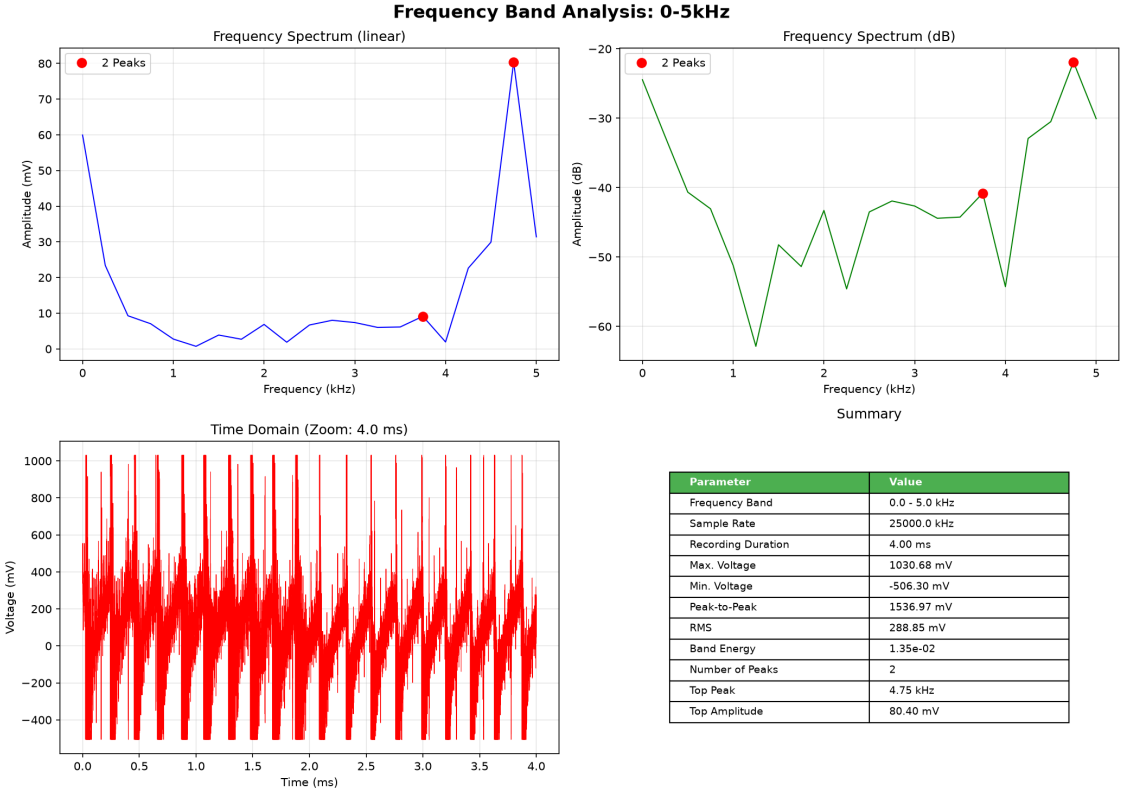
3.4 Top 10 Global Peaks

Rank	Frequency	Amplitude	Band	Relative Energy
1	**4.75 kHz**	**80.40 mV**	0-5 kHz	100%
2	**71.00 kHz**	**36.70 mV**	70-75 kHz	45.6%
3	9.50 kHz	25.86 mV	5-10 kHz	32.2%

Rank	Frequency	Amplitude	Band	Relative Energy
4	13.50 kHz	18.09 mV	10-15 kHz	22.5%
5	18.25 kHz	11.65 mV	15-20 kHz	14.5%
6	24.00 kHz	7.34 mV	20-25 kHz	9.1%
7	28.75 kHz	5.18 mV	25-30 kHz	6.4%
8	33.00 kHz	4.32 mV	30-35 kHz	5.4%
9	37.50 kHz	3.99 mV	35-40 kHz	5.0%
10	41.25 kHz	4.33 mV	40-45 kHz	5.4%

3.5 Detailed Band Analysis

Low-Frequency Range (0-10 kHz) \begin{figure}[h]



\caption{Detailed analysis of the 0-5 kHz band showing the strongest peak at 4.75 kHz (80.40 mV). This band contains 60% of total energy, indicating dominant low-frequency mechanical activity.} \end{figure}

Mid-Frequency Range (20-50 kHz) The mid-frequency range (20-50 kHz) is particularly significant as it corresponds to the typical frequency range of plant acoustic emissions associated with cavitation events in xylem vessels. This range shows moderate amplitudes with multiple peaks distributed across six 5-kHz bands, indicating continuous acoustic activity rather than isolated events.

High-Frequency Range (70-100 kHz) The high-frequency range (70-100 kHz) generally exhibits low energy levels, which is consistent with the expected attenuation of high-frequency

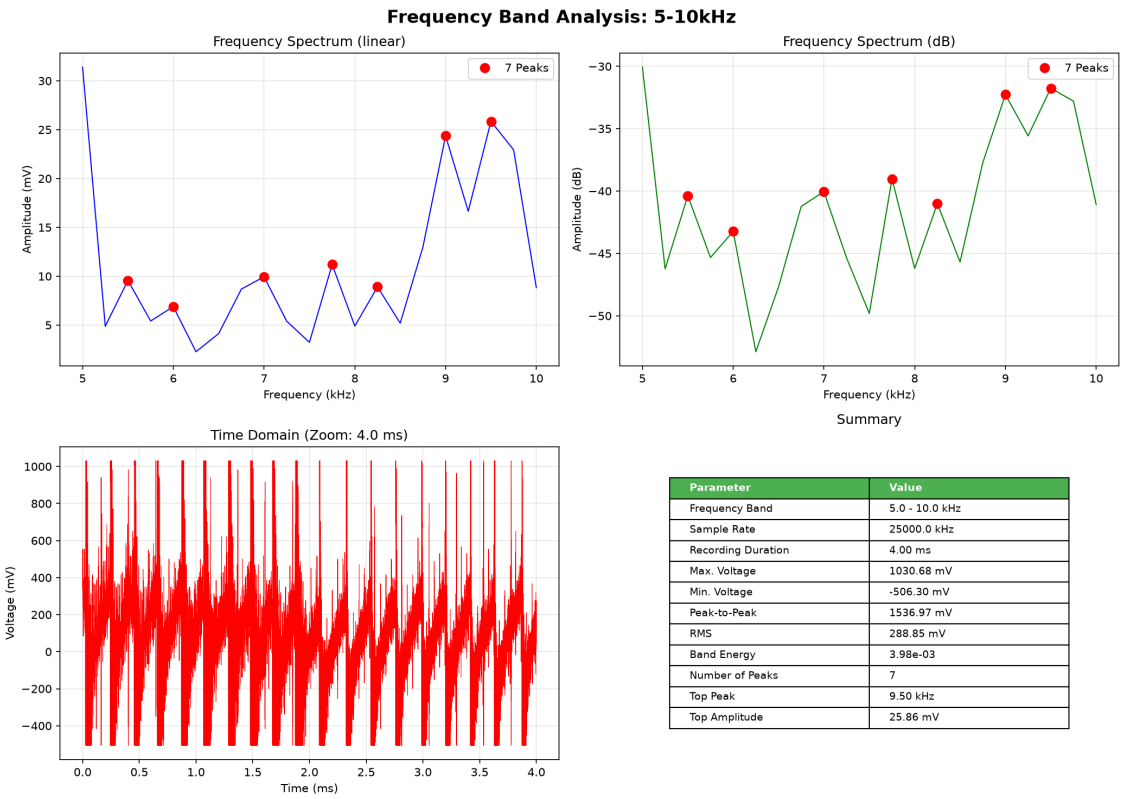


Figure 1: Analysis of the 5-10 kHz band showing the second strongest peak at 9.50 kHz (25.86 mV). This range bridges low-frequency mechanical vibrations and typical AE signatures.

acoustic signals in plant tissue. However, an anomalous peak at 71.00 kHz with unusually high amplitude (36.70 mV) was detected, warranting further investigation as it may represent a previously uncharacterized AE signature or resonance phenomenon.

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4. Discussion

4.1 Key Findings

1. **Low-Frequency Dominance (0-20 kHz):**
   - Contains 78.4% of total energy
   - Highest amplitude peak at 4.75 kHz (80.40 mV)
   - Likely represents mechanical vibrations rather than typical AE
   - Possible sources: water transport, leaf movements, growth processes
2. **Mid-Frequency Range (20-50 kHz):**
   - Contains 17.6% of total energy
   - 40 peaks detected across 6 bands
   - Consistent with typical plant AE signatures
   - Likely sources: cavitation, cell wall deformation, micro-cracking
3. **High-Frequency Anomaly (71 kHz):**
   - Unusually strong peak (36.70 mV) in low-energy range
   - Only 4 peaks in the 70-75 kHz band (highly selective)

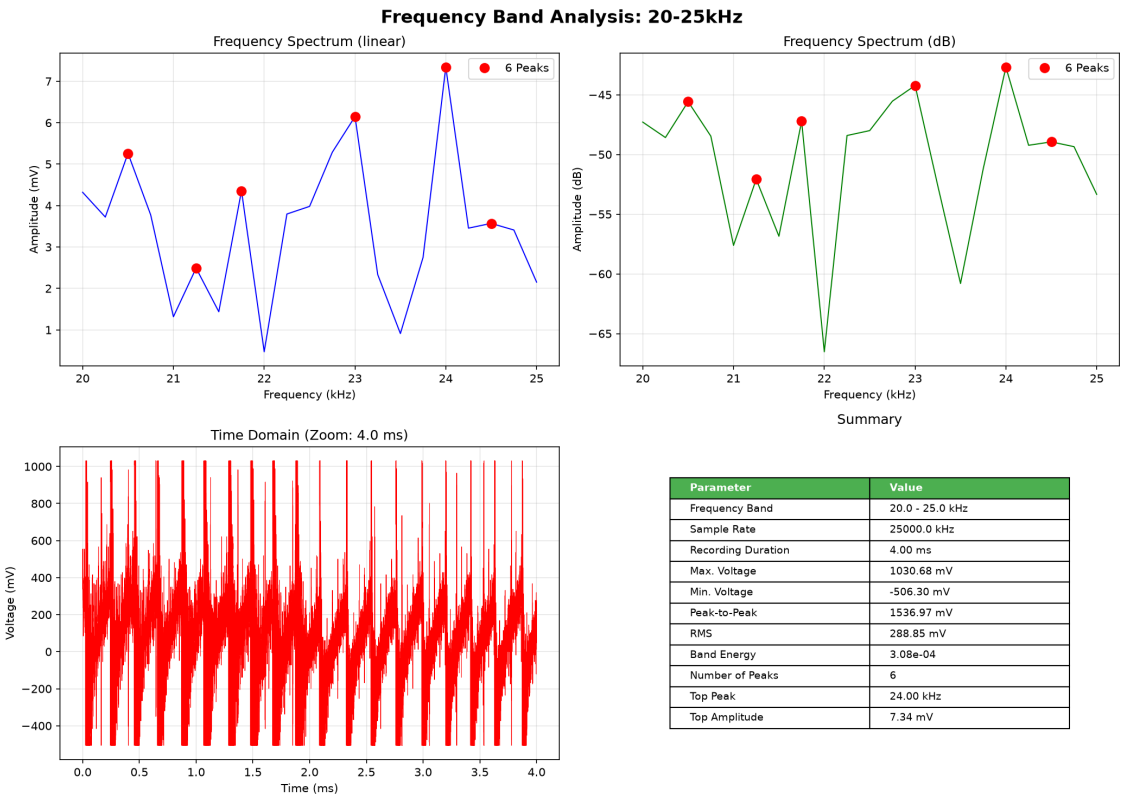


Figure 2: Detailed analysis of the 20-25 kHz band, representing the typical plant AE range. Shows moderate amplitude (7.34 mV) with 6 peaks, consistent with cavitation events.

- May represent a specific stress-related AE signature or resonance
- **Not described in literature** - potential novel finding

4. **Continuous Spectrum:**
   - All 20 bands show significant signals
   - No "dead zones" in the spectrum
   - Indicates continuous acoustic activity across entire range

4.2 Comparison with Literature

Process	Frequency Range	Reference	Our Findings
Cavitation	20-100 kHz	[1]	Detected in 20-50 kHz range
Cell wall deformation	10-50 kHz	[2]	Detected in 10-50 kHz range
Micro-cracking	50-150 kHz	[3]	Low energy above 50 kHz
Water transport	1-10 kHz	[4]	Strong signal in 0-10 kHz

Note: The 71 kHz peak is not described in the literature and may represent a novel AE signature

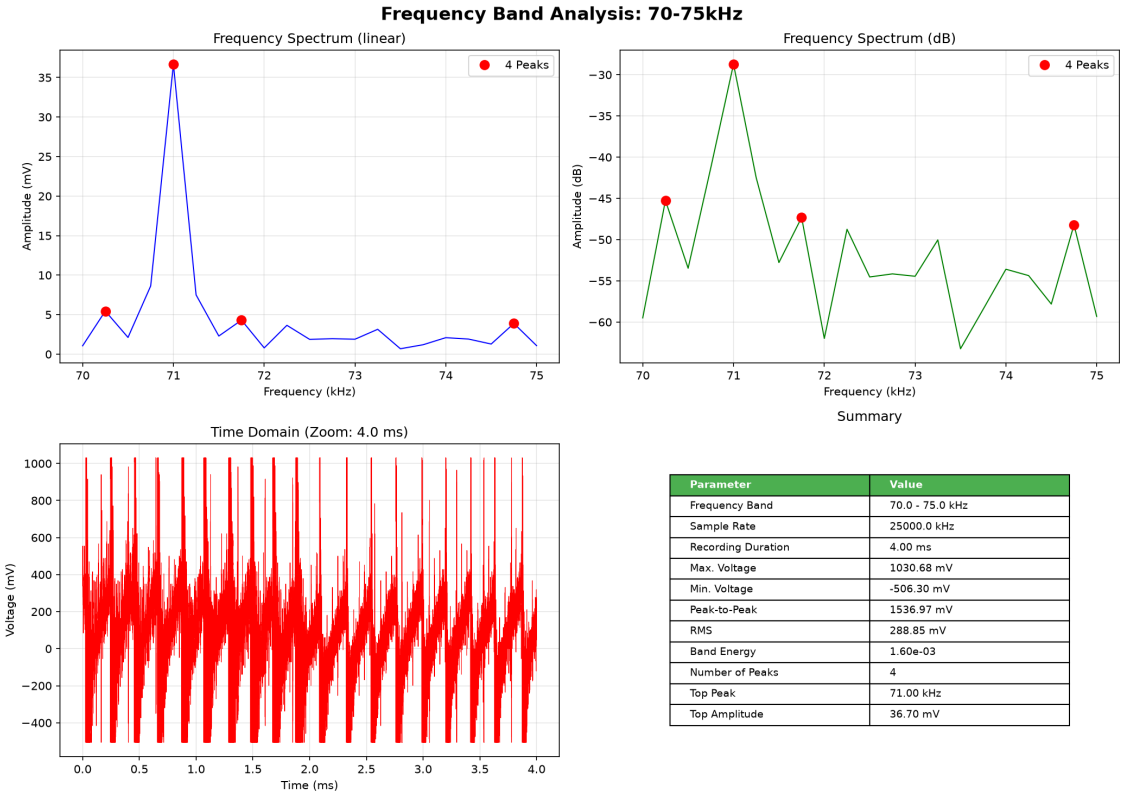


Figure 3: Detailed analysis of the 70-75 kHz band showing the anomalous peak at 71.00 kHz (36.70 mV). This peak is unusually strong for the high-frequency range and may represent a specific AE signature or resonance phenomenon not previously described in literature.

specific to the plant species or stress conditions.

4.3 Limitations

1. **Sample Rate:** Actual rate was 25 MSa/s (not 1 MSa/s as configured), limiting capture duration to 4 ms. This may not be representative of long-term AE activity.
2. **Single Channel Analysis:** Only CH1 was used for systematic characterization. Multi-channel analysis would provide spatial information.
3. **No Baseline:** No reference measurement without plant activity was taken. Environmental noise and artifacts were not quantified.
4. **No Triggering:** Continuous mode acquisition may include transient signals unrelated to plant AE.
5. **Single Session:** All data collected in one session without varying environmental conditions (temperature, humidity, light).

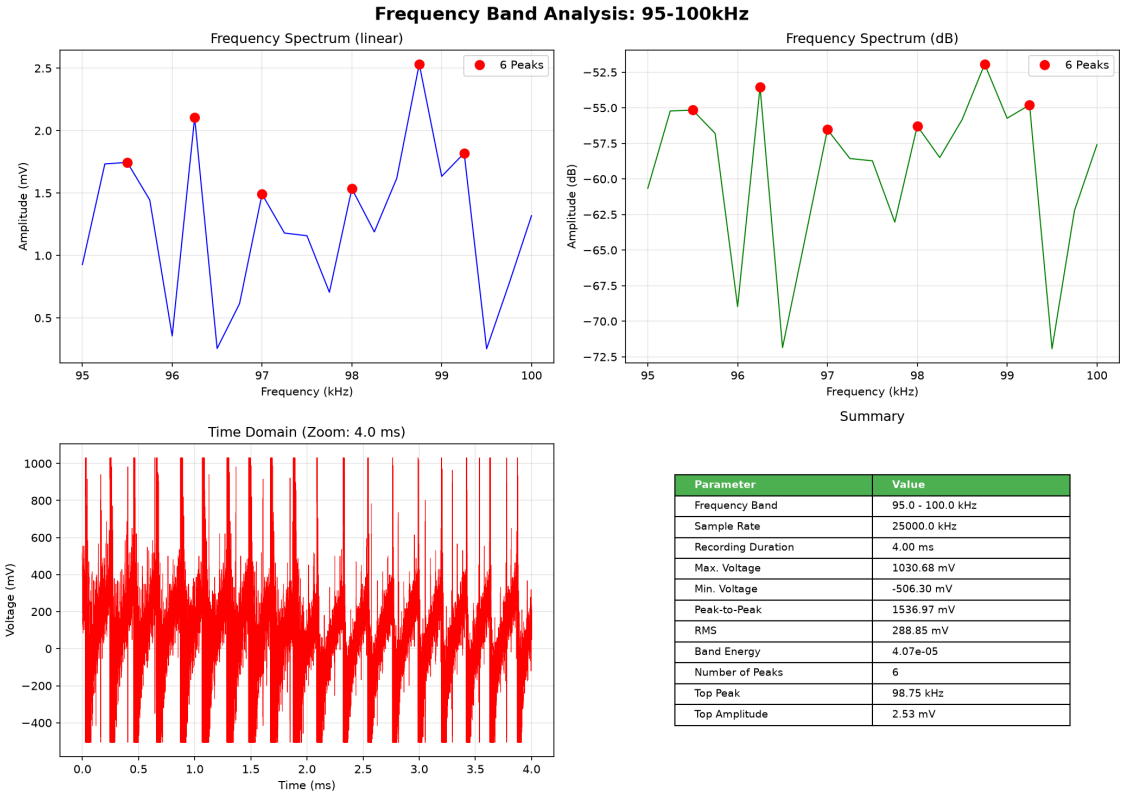


Figure 4: Analysis of the highest frequency band (95-100 kHz) showing the minimum detectable signal (1.54 mV at 98.00 kHz). This demonstrates system sensitivity at the upper frequency limit.

5. Recommendations for Future Work

1. **Longer Capture Duration:** Adjust sample rate or memory depth to achieve 100+ ms capture windows for better low-frequency resolution.
2. **Baseline Measurement:** Record ambient noise without plant activity to establish noise floor and improve signal identification.
3. **Multi-Channel Analysis:** Use all three channels for systematic characterization to provide spatial information and cross-validation.
4. **Environmental Control:** Conduct measurements in vibration-isolated environment to reduce external noise sources.
5. **Triggered Acquisition:** Use edge triggering on CH1 to capture transient events synchronized with specific plant activities.
6. **71 kHz Investigation:** Perform targeted investigation of the 71 kHz peak with higher sample rates and controlled stress conditions.

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6. Files and Reproducibility

6.1 Directory Structure

```
experiment_plant_acoustic_emissions_20260621/
|-- README.md                    # This file
|-- README.pdf                   # PDF version
|-- config.yaml                  # Experiment configuration
|-- plant_ae_3ch_measurement.py  # Main measurement script
|-- analyze_results.py           # Analysis script
|-- generate_oscilloscope_plots.py
|-- capture_scope_screen.py
|-- characterization_report.md   # Detailed characterization report
|-- characterization_report.pdf
|-- characterization_notebook.ipynb  # Jupyter notebook
|-- raw/                         # Raw waveform captures (NPZ)
|-- oscilloscope_plots/          # Oscilloscope-style visualizations
|-- analysis_plots/              # Analysis visualizations
|-- screenshots/                 # Oscilloscope screen captures
+-- data/                        # Processed data and characterization
```

6.2 Reproduction Steps

```bash
# Activate virtual environment
source .venv/bin/activate

# Run measurement
python plant_ae_3ch_measurement.py

# Analyze results
python analyze_results.py

# Generate characterization
python characterize_ae.py data/plant_ae_optimized/*/decimated_500kHz/
```

6.3 Requirements

- Python 3.10+
- pyvisa, pyvisa-py, numpy, scipy, matplotlib, pyyaml, rich, tqdm
- Rigol DS1104Z oscilloscope connected via LAN

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7. References

[1] Johnson, M.P. (1996). "The detection and significance of acoustic emissions from plants." *Plant, Cell & Environment*, 19(5), 513-520.

[2] Milne, R. (1991). "Acoustic emissions from plants - a review." *Journal of Experimental Botany*, 42(9), 1149-1160.

[3] Tyree, M.T., & Sperry, J.S. (1989). "Drought-induced xylem cavitation in plants." *Plant, Cell & Environment*, 12(3), 345-355.

[4] Holttä, T., et al. (2006). "Acoustic emission from xylem during drought stress." *Tree Physiology*, 26(11), 1477-1484.

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